

Stanley Okwii

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Summary

Senior software engineer with 6+ years building production systems in Python and TypeScript on AWS, across payments, analytics products and data pipelines. Currently founding engineer on an AI research assistant, owning the hybrid retrieval pipeline, agent orchestration, and multi-tenant cloud infrastructure. Experienced across the full lifecycle—from schema design and IaC to React interfaces—leveraging agentic development harnesses for rapid delivery while maintaining strict ownership of architecture, security, and system reliability.

Skills

Programming: Python, TypeScript, JavaScript, SQL, Dart

AI/ML: RAG pipelines, LLM integration (Claude and Gemini APIs), prompt engineering, agent and tool use workflows, embeddings, vector search (pgvector), hybrid retrieval

Backend and Frameworks: Django, FastAPI, Flask, Node.js, Express, GraphQL, Celery, Redis

Frontend and Mobile: React, Next.js, Flutter

Databases: PostgreSQL, MongoDB, Redis, MySQL

Infrastructure: AWS (ECS, Bedrock, CloudFormation), GCP, Docker, GitHub Actions, Supabase, Sentry, New Relic, Terraform, Kubernetes

Testing: Jest, Playwright, Pytest

Developer Tooling: Claude Code (custom skills and agent workflows), Dev Containers, CI/CD

Experience

Founding Engineer, Access to Energy – London, UK Jan 2026 – present

An AI research assistant that answers questions on energy access in Africa, citing evidence from a curated library of sector reports. Team of 4. I own the backend, the AI pipeline and the infrastructure.

- Built the core hybrid search engine combining keyword matching (BM25) and semantic vector search (768-dim Gemini embeddings, HNSW) in PostgreSQL, delivering accurate passage retrieval without external database overhead.
- Built the end-to-end RAG assistant in Python (bespoke pipeline without LangChain bloat), routing questions to AWS Bedrock (Claude 3 Haiku) with streaming inline citations and automated Gemini fallback.
- Built the background document ingestion pipeline using a custom Redis Streams worker (consumer groups, dead-letter retries) to parse, chunk, and embed large PDFs without degrading API response times.
- Designed multi-tenant data isolation and access controls in PostgreSQL, letting multiple client organisations share one deployment while keeping their documents fully partitioned and metering usage.
- Automated AWS infrastructure provisioning using modular nested CloudFormation stacks, deploying ECS Fargate microservices, ElastiCache Redis 7.1, ALB, and Fluent Bit telemetry to Axiom.

Full Stack Software Engineer (Contract, Part Time), Hoppers – Remote Oct 2025 – present

Event ticketing platform (hoppers.ug). Built the multi-provider payment engine and high-concurrency reservation system.

- Built the multi-provider payment system (DPO, IoTec), implementing idempotent webhook handling via row-level locking (SELECT FOR UPDATE) to eliminate duplicate charges.
- Prevented ticket overselling during demand spikes by building a high-concurrency reservation engine with guarded atomic updates (Django F expressions) and database check constraints.
- Enforced purchase limits and venue table reservations by serializing booking state transitions with row-level locks, stopping race conditions during high-traffic checkout.
- Delivered core platform features end-to-end across the Django backend, React admin panel, and mobile app, covering QR code ticket verification and role-based permissions.
- Created an automated testing harness using Claude Code agents to turn GitHub issues into draft pull requests verified by Playwright UI diffs and replayed API test suites.

Senior Full Stack Software Engineer, Harver – New York, USA (remote) Mar 2023 – May 2025

Promoted to Senior in March 2023. Stepped away in May 2025 to complete MSc in Data Analytics full time.

- Owned the architecture and delivery of three enterprise client products—candidate assessment matching, recruiter portal, and analytics dashboards—built with Django REST Framework, PostgreSQL, and React.
- Led technical design and RFCs across engineering, product, and design, building asynchronous task queues with Celery and RabbitMQ to process candidate scoring pipelines.
- Directed Tier-3 incident response for enterprise clients, isolating distributed regressions via New Relic APM and Sentry traces to deliver permanent root-cause fixes.
- Ran cross-functional project planning, breaking complex product roadmaps into phased milestones and managing risk mitigation to hit scheduled release dates.
- Conducted 15+ technical interviews for system design and code quality, and authored onboarding documentation for core services to accelerate new hire ramp-up.

Full Stack Software Engineer, Harver – New York, USA (remote) Sept 2019 – Mar 2023

- Introduced structured logging and APM monitoring across distributed Django services with Sentry and New Relic, replacing speculative debugging with direct error-to-cause tracing.
- Built talent mobility dashboards in React, enabling enterprise clients to track employee career progression and skills across their organizations.
- Profiled high-latency API endpoints, eliminating N+1 database queries and implementing Memcached caching on hot data paths.
- Automated deployment pipelines with GitHub Actions and Docker, replacing manual deployment procedures with one-click CI/CD releases.

Mentorship

CareerFoundry (Online Coding Bootcamp) – Berlin, Germany Mar 2023 – Jan 2026

Full Stack Web Development Mentor (Freelance)

- Mentored 23+ students through the Full Stack Web Development program and held a 5/5 student rating for the quality and speed of my feedback.

Education

MSc Birkbeck, University of London, Data Analytics – London, UK Oct 2024 – Sept 2025

- Merit. Dissertation on adversarial robustness in image classification: implemented PGD attacks from scratch and adversarially retrained ResNet50 on an ImageNet subset.

BSc Makerere University, Software Engineering – Kampala, Uganda Aug 2014 – Sept 2018

- 2:1 (Upper Second Class Honours)